	pass@k		
	$k = 1$	$k = 10$	$k = 100$
GPT-Neo 1.3B	4.79%	7.47%	16.30%
GPT-Neo 2.7B	6.41%	11.27%	21.37%
GPT-J 6B	11.62%	15.74%	27.74%
GPT-NeoX 20B	15.4%	25.6%	41.2%
Codex-300M	13.17%	20.37%	36.27%
Codex-679M	16.22%	25.7%	40.95%
Codex-2.5B	21.36%	35.42%	59.5%
Codex-12B	28.81%	46.81%	72.31%
BLOOM-560M	0.82%	3.02%	5.91%
BLOOM-1.1B	2.48%	5.93%	9.62%
BLOOM-1.7B	4.03%	7.45%	12.75%
BLOOM-3B	6.48%	11.35%	20.43%
BLOOM-7.1B	7.73%	17.38%	29.47%
BLOOM	15.52%	32.20%	55.45%
BLOOMZ-560M	2.18 %	4.11%	9.00%
BLOOMZ-1.1B	2.63%	6.22%	11.68%
BLOOMZ-1.7B	4.38%	8.73%	16.09%
BLOOMZ-3B	6.29%	11.94%	19.06%
BLOOMZ-7.1B	8.06%	15.03%	27.49%
BLOOMZ	12.06%	26.53%	48.44%

Table 9: Performance on HumanEval (Chen et al., 2021). Non-BLOOM results come from prior work (Chen et al., 2021; Fried et al., 2022). The Codex model is a language model that was finetuned on code, while the GPT models (Black et al.; Wang and Komatsuzaki, 2021; Black et al., 2022) are trained on a mix of code and text like BLOOM.

sentence completion (XCOPA and XStoryCloze). After going through multilingual multitask finetuning (BLOOMZ), zero-shot performance significantly improves on the depicted held-out tasks. Despite also being multitask finetuned, T0 performs badly on the multilingual datasets shown due to it being a monolingual English model. Additional results provided in Muennighoff et al. (2022b), however, show that models finetuned on xP3 also outperform T0 on English datasets when controlling for size and architecture. This is likely due to T0's finetuning dataset (P3) containing less diverse datasets and prompts than xP3. Multitask finetuning performance has been shown to correlate with the amount of datasets and prompts (Chung et al., 2022).

4.8 Embeddings

In Section 3.5, we have outlined the contrastive finetuning procedure for creating SGPT-BLOOM text embedding models. In Table 10, we report benchmarking results on two

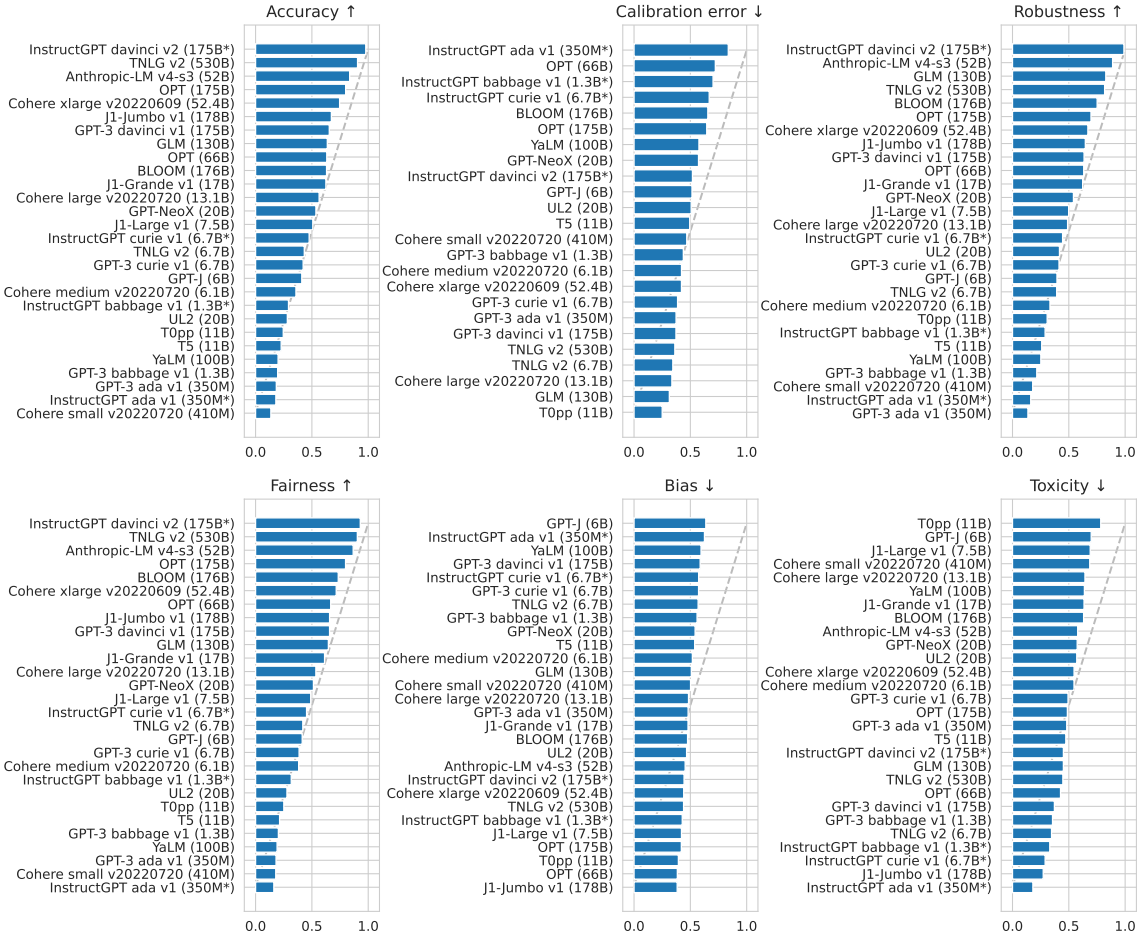

Figure 10: Results for a wide variety of language models on the 5-shot HELM benchmark. Taken from Liang et al. (2022)

multilingual datasets from the Massive Text Embedding Benchmark (MTEB, Muennighoff et al., 2022a). We find that SGPT-BLOOM-7.1B-msmarco[36] provides state-of-the-art performance on several classification and semantic textual similarity splits. However, with 7.1 billion parameters it is an order of magnitude larger than models like the displayed multilingual MiniLM[37] and MPNet[38]. SGPT-BLOOM-1.7B-nli[39] performs significantly worse, likely due to less parameters and its finetuning being shorter (NLI is a much smaller dataset than MS-MARCO). Apart from the BLOOM models, ST5-XL[40] is the largest model with 1.2 billion parameters. However, as an English-only model its performance on non-English

36. `hf.co/bigscience/sgpt-bloom-7b1-msmarco`
37. `hf.co/sentence-transformers/paraphrase-multilingual-MiniLM-L12-v2`
38. `hf.co/sentence-transformers/paraphrase-multilingual-mpnet-base-v2`
39. `hf.co/bigscience/sgpt-bloom-1b7-nli`
40. `hf.co/sentence-transformers/sentence-t5-xl`

Figure 11: BLOOMZ zero-shot task generalization. Five untuned prompts are evaluated for each dataset and plotted. T0 is monolingual (English) while other models are multilingual. T0 performance may be hurt by its inability to tokenize some non-English texts.

languages is poor. The languages displayed are part of the BLOOM pretraining corpus. Performance on more languages and datasets can be inspected on the MTEB leaderboard[41].

4.9 Multilingual Probing

Probing has emerged as a significant evaluation paradigm to analyze and interpret the inner workings of LLMs (Ettinger et al., 2016; Adi et al., 2017; Belinkov et al., 2017; Hupkes et al.,

41. `hf.co/spaces/mteb/leaderboard`